

Lemierre Syndrome: A Comprehensive Disease Characteristics Report

Disease: Lemierre Syndrome (LS) **MONDO ID:** MONDO:0015306 **Category:** Infectious Disease
Report type: Disease knowledge-base entry (multi-iteration autonomous literature synthesis)
Evidence base: 27 PubMed sources; predominantly human clinical case series, systematic reviews, and national registry studies, supplemented by veterinary/in-vitro mechanistic studies. No patient-level data files provided.

Summary

Lemierre syndrome is a rare, life-threatening, acquired infectious disease defined by septic thrombophlebitis of the internal jugular vein (IJV) that develops as a complication of an oropharyngeal (usually pharyngotonsillar) infection, with subsequent metastatic septic emboli that spread most often to the lungs. It is caused predominantly by the Gram-negative obligate anaerobe *Fusobacterium necrophorum*, which is identified in roughly 78% of cases in systematic reviews. The classic causal triad is: (1) oropharyngeal infection → (2) IJV septic thrombophlebitis → (3) metastatic septic emboli. The syndrome characteristically strikes previously healthy adolescents and young adults (median age ~15–20 years) with a male predominance.

Critically, Lemierre syndrome is NOT a genetic disease. No causal human gene, Mendelian inheritance pattern, penetrance/expressivity behavior, heritable variant, or founder effect exists for this condition. It is an acquired complication of bacterial infection. Consequently, the many sections of the research template dealing with causal genes, pathogenic variants, ACMG variant classification, gnomAD allele frequencies, genetic testing (WGS/WES/panels/karyotyping), carrier screening, genetic counseling, and inheritance are **Not Applicable**, and this report explicitly documents them as such rather than fabricating content. The etiologically relevant "genome" here is that of the pathogen: the *F. necrophorum* leukotoxin operon and associated virulence factors.

Prognosis is favorable with prompt treatment but the illness remains serious. Modern mortality is approximately 5% (versus up to 80% in the pre-antibiotic era, when death commonly occurred within 7–15 days), but complication rates are high—sepsis in ~83%, thrombocytopenia in ~75%, ICU admission in ~43%, and septic shock in ~18% of hospitalized patients. Management centers on prolonged anaerobe-active antibiotic therapy (a beta-lactam/beta-lactamase inhibitor or third-generation cephalosporin combined with metronidazole, or a carbapenem) plus source control; adjunctive anticoagulation remains controversial and of unproven benefit. With appropriate therapy the disease is self-limited/curable (no chronic or relapsing course), and thrombus resolves or recanalizes in ~78% of cases on follow-up imaging. A key evidence gap is that the platelet-aggregation/thrombus-initiation step of human pathophysiology is inferred from in-vitro and ruminant data rather than directly demonstrated in humans.

Key Findings

Finding 1 — Definition and dominant pathogen

Lemierre syndrome is septic thrombophlebitis of the internal jugular vein caused predominantly by *Fusobacterium necrophorum* following an oropharyngeal infection. A systematic review of 143 pediatric cases found *F. necrophorum* in 78.3% of patients, with fever as the most common presenting symptom (92%); CT imaging was used in 89.5% of cases and was most useful for detecting IJV thrombosis and septic emboli (PMID: 42127655): "*F. necrophorum was the predominantly identified organism in 78.3% of patients.*" The defining pathology is captured directly in the literature: "*It is characterized by septic thrombophlebitis of the internal jugular vein and subsequent metastatic abscess formation. The most common causative pathogen of LS is Fusobacterium necrophorum*" (PMID: 41517763). The classic clinical sequence—oropharyngeal infection → IJV septic thrombophlebitis → metastatic septic emboli (most often to the lung)—organizes the entire disease concept.

Ontology anchors: MONDO:0015306 (Lemierre syndrome); UBERON:0001584 (internal jugular vein); NCBI:txid859 (*Fusobacterium necrophorum*); HP:0001945 (Fever).

Finding 2 — Epidemiology: rare, rising, young, male-predominant

Lemierre syndrome is rare but with rising incidence, currently estimated at roughly 1–10 cases per million per year (a commonly cited central estimate is ~3.6 per million). A Swedish nationwide study (2010–2017, n = 300 invasive *F. necrophorum* infections) documented that the incidence of invasive *F. necrophorum* infection increased from 2.9 to 5.0 cases/million/year ($p = 0.001$), that 104/300 (35%) of these patients developed Lemierre syndrome, and that the median age of LS patients was 20 years (PMID: 31843654): *"The incidence increased from 2.9 to 5.0 cases/million/year from 2010-13 to 2014-17 ($p = 0.001$)."* A population-level incidence estimate of *"3.6 cases per million population"* is reported in PMID: 42576194. The disease has been termed "the forgotten disease" because it became rare after the introduction of antibiotics but has been re-emerging since the 2000s. It preferentially affects previously healthy adolescents and young adults with a male predominance.

Ontology anchors: age of onset — adolescent/young adult (HP:0011462, Young adult onset).

Finding 3 — Severe systemic illness; ~5% modern mortality

Lemierre syndrome causes severe systemic illness with high complication rates. In the Swedish cohort of 104 LS patients, 75% (72/96) had thrombocytopenia on admission, 83% (86/104) had sepsis, 18% (19/104) developed septic shock, and 43% (45/104) required intensive care (PMID: 31843654): *"72/96 (75%) had thrombocytopenia on admission, 86/104 (83%) had sepsis, 19/104 (18%) developed septic shock and 45/104 (43%) needed intensive [care]."* Modern mortality is approximately 5%, rising with diagnostic delay, whereas in the pre-antibiotic era the disease *"was often characterized by a fatal course within 7-15 days with a mortality rate that could reach up to 80% of cases"* (PMID: 40265142). Long-term complications were reported in 4.2% of pediatric cases (PMID: 42127655).

Complication	Frequency (Swedish LS cohort, n=104)
Sepsis	83% (86/104)
Thrombocytopenia on admission	75% (72/96)
ICU admission	43% (45/104)
Septic shock	18% (19/104)

Ontology anchors: HP:0100806 (Sepsis); HP:0001873 (Thrombocytopenia); HP:0031273 (Septic shock).

Finding 4 — Leukotoxin is the major virulence factor

F. necrophorum leukotoxin is the major virulence factor driving Lemierre pathophysiology, with endotoxin (LPS), hemolysin, hemagglutinin, proteases, and adhesins contributing. Reviews conclude that *"leukotoxin and endotoxin are believed to be more important than other toxins in overcoming the host's defence mechanisms to establish the infection"* (PMID: 8711893). Mechanistically, purified leukotoxin activates polymorphonuclear leukocytes (PMNs) and induces their apoptosis/necrosis: *"The ability of F. necrophorum leukotoxin to modulate the host immune system by its toxicity, including cellular activation of PMNs and apoptosis-mediated killing of phagocytes and immune effector cells, represents a potentially important mechanism of its pathogenesis"* (PMID: 12117974). Subspecies *necrophorum* (biotype A) is more virulent than subspecies *funduliforme* (biotype B) (PMID: 16701574).

Ontology anchors: GO:0006954 (inflammatory response); GO:0006915 (apoptotic process); GO:0090729 (toxin activity); CL:0000775 (neutrophil); CHEBI:16412 (lipopolysaccharide).

Finding 5 — Diagnosis: contrast-enhanced CT + anaerobic blood cultures + molecular methods

Diagnosis relies on contrast-enhanced CT of the neck and chest (demonstrating IJV thrombosis and septic emboli) plus anaerobic blood cultures, with molecular methods increasingly used. In the pediatric systematic review, *"Computed tomography (CT) imaging was utilized in 89.5% of cases, and was most useful for detecting thrombosis of the internal jugular vein and septic emboli"* (PMID: 42127655). The two-pronged confirmatory standard is stated directly: *"The diagnosis of Lemierre's syndrome is typically confirmed through the identification of thrombophlebitis of the internal jugular vein on radiographic imaging and the isolation of anaerobic bacteria in blood cultures"* (PMID: 39840169). Contrast-enhanced cervicothoracic CT is the workhorse (PMID: 40224244). Because anaerobic cultures may be slow or negative, molecular diagnostics—16S rDNA sequencing/targeted PCR (PMID: 31843654) and targeted next-generation sequencing combined with metagenomic capture—enable rapid *F. necrophorum* detection (PMID: 41517763). Common laboratory abnormalities include thrombocytopenia, elevated CRP, and leukocytosis.

Ontology anchors: diagnostic imaging — contrast-enhanced CT; LOINC-mappable labs (CRP, platelet count, WBC); anaerobic blood culture.

Finding 6 — Treatment: prolonged anaerobe-active antibiotics + source control; anticoagulation controversial

Treatment centers on prolonged anaerobe-active antibiotics with source control. First-line therapy is described as: *"Immediate treatment involves broad-spectrum antibiotic therapy, often utilizing a third-generation cephalosporin or a beta-lactam in combination with metronidazole"* (PMID: 40224244). Carbapenems are also reported as a good therapeutic choice (PMID: 20570017). Anticoagulation remains controversial: a systematic review found parenteral anticoagulation used initially in 12/14 patients and DOAC outcomes similar to warfarin, but concluded that *"the thromboembolic events have rarely led to significant complications; thrombi typically resolve independently, and concerns for bleeding risks are well founded"* (PMID: 32909436). Anticoagulation was used in 62.2% of pediatric cases and surgery in 6% (PMID: 42127655). Surgical options include abscess drainage, with IJV ligation/excision reserved for non-responders (PMID: 39257960).

Treatment element	Detail	NCIT-type mapping
Beta-lactam/BLI (e.g., ampicillin-sulbactam, piperacillin-tazobactam)	Empiric anaerobe coverage	Antibacterial agent
Third-gen cephalosporin + metronidazole	Common first-line combination	Metronidazole (NCIT antibacterial)
Carbapenem (e.g., meropenem)	Alternative broad-spectrum choice	Carbapenem antibiotic
Anticoagulation (LMWH/DOAC/warfarin)	Selected high-risk cases; controversial	Anticoagulant therapy
Abscess drainage / IJV ligation-excision	Source control; salvage for non-responders	Surgical procedure

Ontology anchors: CHEBI:6077 (metronidazole); NCIT antibacterial/anticoagulant/surgical-procedure branches.

Finding 7 — Anatomy and phenotypes

The primary source is oropharyngeal (tonsillitis/pharyngitis, peritonsillar abscess), with odontogenic and otogenic (otitis media/mastoiditis) sources also described (PMID: 42517948, PMID: 12109395). Septic thrombophlebitis localizes to the internal jugular vein, and septic emboli disseminate preferentially to the lungs (pulmonary emboli/cavitary consolidations); in

one series *"All [patients had] sore throat and pulmonary embolisms"* (PMID: 20570017). Metastatic and contiguous spread can reach the intracranial venous sinuses, orbit, joints, and pleura (empyema), and can be complicated by DIC and ARDS. An otogenic pediatric series documented that *"Internal jugular vein thrombosis occurred in 80%, sigmoid sinus thrombosis in 60%, and cavernous sinus thrombosis in 30%"* (PMID: 42141231). Common phenotypes include fever (92%), sore throat, neck pain/swelling, chest pain, hemoptysis, and dyspnea.

Level	Structure	UBERON / ontology term
Primary source	Oropharynx / palatine tonsil	UBERON:0001729 (oropharynx); UBERON:0002373 (palatine tonsil)
Primary vascular lesion	Internal jugular vein	UBERON:0001584
Dominant embolic target	Lung	UBERON:0002048
Intracranial extension	Sigmoid/cavernous venous sinuses	UBERON:0006459 (sigmoid sinus); UBERON:0004024 (cavernous sinus)
Body systems	Cardiovascular, respiratory, digestive/upper aerodigestive	—

Phenotype ontology anchors: HP:0001945 (Fever); HP:0002098 (Respiratory distress/dyspnea); HP:0002105 (Hemoptysis); HP:0100749 (Chest pain).

Finding 8 — Not genetic; expanding etiologic spectrum ("Lemierre-like syndrome")

Lemierre syndrome is not a genetic disease—no causal human gene, inheritance pattern, or heritable variant is described; it is an acquired complication of bacterial infection. Beyond classic *F. necrophorum*, cases arise from viridans group streptococci (often odontogenic; PMID: 42517948) and other anaerobes/streptococci. "Lemierre-like syndrome" (LLS) from non-oropharyngeal sources is increasingly recognized: *"In contrast to classic Lemierre syndrome, sources of infection are not related to oropharyngeal infections, as are frequent soft tissue infections. In recent years, Staphylococcus aureus has been identified as an emergent pathogen that causes this syndrome. The mortality rate of LLS caused by this pathogen is approximately 16%"* (PMID: 38363930). An "incomplete Lemierre's syndrome" variant is also defined: *"Incomplete Lemierre's syndrome is a bacterial oropharyngeal infection, complicated by septic*

emboli to end organs, but without thrombophlebitis of the internal jugular vein" (PMID: 42286749). Risk factors are environmental/infectious: preceding pharyngotonsillitis, young age, and restricted/inappropriate antibiotic use for sore throat (PMID: 12109395).

Finding 9 — Natural animal disease and models (necrobacillosis)

F. necrophorum causes major naturally occurring disease in animals (necrobacillosis), which informs pathogenesis and prevention. It is a normal alimentary-tract commensal and opportunistic pathogen; *"Of these, bovine liver abscesses and foot rot are of significant concern to the cattle industry"* (PMID: 16701574). Liver abscesses arise when ruminal organisms enter the portal circulation secondary to ruminal acidosis. A randomized, blinded feedlot field trial demonstrated that a *F. necrophorum* bacterin reduced liver abscesses (odds ratio 0.27, $P = 0.05$) and footrot (OR 0.18, $P = 0.03$) in forage-fed cattle: *"The odds that a vaccinated animal in the ALF group would have an A or A+ liver abscess at slaughter were less than 1/3 the odds that an unvaccinated animal"* (PMID: 16363327). Isolates have also been recovered from the respiratory tract of white-tailed deer (*Odocoileus virginianus*) (PMID: 24590666). Leukotoxin virulence is studied primarily in bovine leukocyte/PMN in-vitro models (PMID: 12117974).

Taxonomy anchors: *Bos taurus* (NCBI:txid9913, cattle); *Odocoileus virginianus* (NCBI:txid9874, white-tailed deer); *Fusobacterium necrophorum* (NCBI:txid859).

Finding 10 — Prevention is secondary; no human vaccine

Prevention of Lemierre syndrome is secondary—early recognition and appropriate anaerobe-active antibiotic treatment of *F. necrophorum* pharyngitis—because no human vaccine or established primary chemoprophylaxis exists. Cost-effectiveness modeling suggests that *"examining throat swabs from 15- to 24-year-olds for F. necrophorum followed by antibiotic treatment will probably be less costly than most other life-saving medical interventions, with a median cost of US\$8,795 per QALY saved"* (PMID: 22886057); only a 20–25% reduction in LS/PTA incidence would be required for cost-effectiveness. The main preventable drivers are diagnostic delay and inadequate antibiotic coverage; restricted or inappropriate antibiotic use (macrolides, narrow cephalosporins, to which *F. necrophorum* may be resistant) for pharyngitis is linked to rising incidence (PMID: 12109395, PMID: 20570017). Early radiographic recognition of impending LS enables timely intervention (PMID: 39840169).

Finding 11 — Temporal course: acute onset, rapid progression, self-limited with treatment

The temporal course is acute/subacute onset days after pharyngitis, rapid progression to sepsis, but self-limited with treatment (no chronic/relapsing course). The classic sequence is antecedent sore throat/pharyngitis, then within roughly one week the development of IJV thrombophlebitis and septic emboli. Imaging can capture striking speed: *"Within four days, the infection rapidly progressed to complete occlusion of the internal jugular vein with pulmonary septic emboli"* (PMID: 39840169). The pre-antibiotic natural history was often fatal within 7–15 days (PMID: 40265142). With appropriate therapy the illness is self-limited/curable—there is no lifelong or relapsing course—though recovery requires prolonged (weeks-long) antibiotics; thrombus resolves or recanalizes in ~78% of cases on follow-up imaging: *"Thrombosis resolved or recanalized in 78% of cases on follow-up imaging"* (PMID: 42141231). Onset is typically in previously healthy adolescents/young adults (median ~15–20 years); it is not congenital.

Finding 12 — Evidence gap: human thrombus-initiation step is inferred

The platelet-aggregation/thrombus-initiation step in human Lemierre pathophysiology is inferred, not directly demonstrated. Reviews list hemagglutinin and platelet-aggregating/hemolytic activity among *F. necrophorum* virulence factors implicated in thrombus formation, but candidly note that *"The pathogenic mechanism of F. necrophorum is complex and not well defined"* (PMID: 16701574, PMID: 8711893). Direct experimental demonstration of *F. necrophorum*-driven platelet aggregation causing IJV thrombosis in humans was not identified in this review; the strongest mechanistic data (leukotoxin-mediated leukocyte apoptosis) derive from ruminant/in-vitro systems (PMID: 12117974). No human transcriptomic/proteomic/metabolomic disease signatures are available.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Oropharyngeal colonization/infection** by *F. necrophorum* (usually acute tonsillitis/pharyngitis or peritonsillar abscess; alternatively odontogenic or otogenic foci) **leads to** local mucosal invasion. (*Well supported* —

P 42127655

P 12109395

2. Local invasion **results in** direct/contiguous spread into the peritonsillar and parapharyngeal soft tissues and the adjacent carotid sheath. *(Well supported anatomically*

P 42517948

3. Bacterial **leukotoxin and endotoxin (LPS)** activate polymorphonuclear leukocytes and then kill phagocytes by apoptosis/necrosis, **enabling immune evasion** and unchecked local proliferation. *(Mechanism demonstrated in vitro/ruminant models —*

P 12117974

P 8711893

4. Endothelial injury plus procoagulant activity (hemagglutinin, platelet-aggregating and hemolytic factors) **is inferred to lead to** septic thrombophlebitis of the internal jugular vein. *(INFERRED in humans — the platelet-aggregation/thrombus-initiation step is not directly demonstrated;*

P 16701574

P 8711893

Finding 12.)

5. IJV septic thrombophlebitis **results in** bacteremia and shedding of infected thrombus fragments → **metastatic septic emboli**, most commonly to the lungs (cavitary consolidations, empyema), and less often to joints, intracranial venous sinuses, and orbit. *(Well supported —*

P 20570017

P 42141231

6. Systemic bacteremia and embolization **lead to** sepsis (~83%), thrombocytopenia (~75%), and, in severe cases, septic shock (~18%), DIC, and ARDS. *(Well supported —*

P 31843654

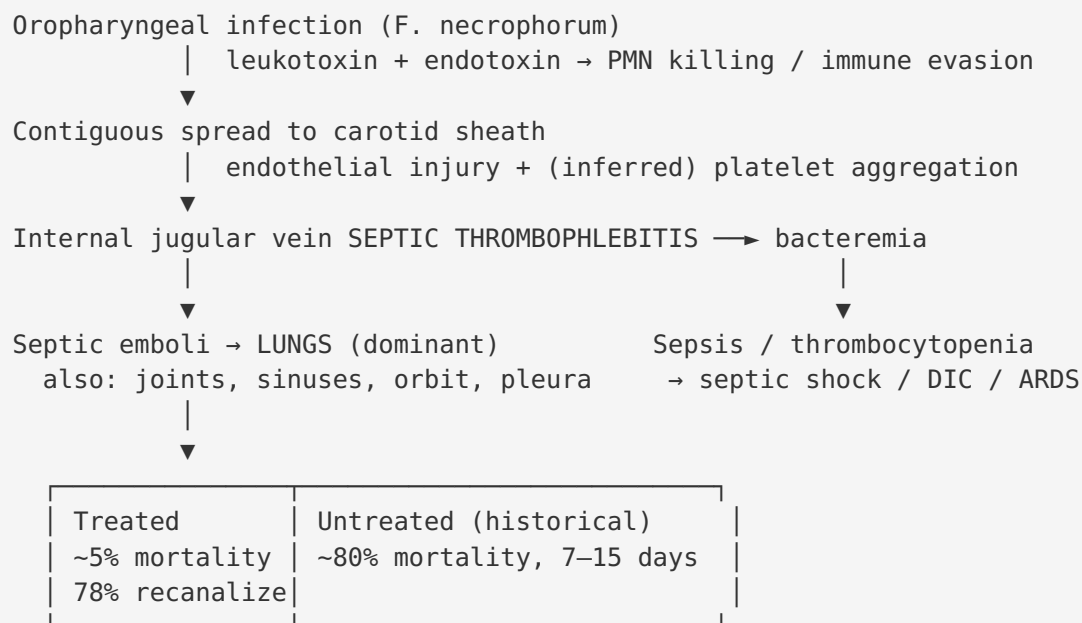
P 42517948

7. **Branch:** with prompt anaerobe-active antibiotics ± source control, the process is **self-limited**, thrombus recanalizes (~78%), and mortality is ~5%; **without** timely therapy, the historical course was fatal within 7–15 days (mortality up to 80%). (*Well supported* —

P 40265142

P 42141231

P 39840169



Upstream vs downstream: The pathogen's leukotoxin/endotoxin-mediated immune subversion is the upstream driver; IJV thrombosis is the pivotal intermediate lesion; septic emboli and multi-organ failure are downstream consequences. **Cell types / processes:** neutrophils (CL:0000775) and other phagocytes are killed by leukotoxin (GO:0006915 apoptotic process; GO:0006954 inflammatory response); vascular endothelium (CL:0000115) and platelets (CL:0000233) participate in thrombus formation (GO:0007596 blood coagulation). Because the human disease is an acquired infection, there are no host causal genes, pathways (Wnt/MAPK/mTOR/PI3K-AKT), epigenetic changes, or heritable protein dysfunctions to annotate; the relevant molecular apparatus is bacterial.

Not Applicable / Not Available Sections (explicit)

Because Lemierre syndrome is an **acquired infectious disease with no genetic etiology**, the following template items are Not Applicable and should be recorded as such in the knowledge base:

Template section	Status	Rationale
Causal genes / OMIM gene entries	Not Applicable	No human causal gene; not Mendelian
Pathogenic variants / ACMG classification / gnomAD allele frequency	Not Applicable	No heritable disease variants
Modifier genes, epigenetic changes, chromosomal abnormalities	Not Applicable	No genetic disease architecture
Inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, carrier frequency	Not Applicable	Non-genetic
Genetic testing (WGS/WES/panels/single-gene/CMA/karyotype/FISH/mtDNA/repeat expansion)	Not Applicable	Diagnosis is microbiological + imaging
Genetic counseling / carrier & prenatal screening	Not Applicable	Non-heritable
Pharmacogenomics	Not Available	No PGx associations described for LS antibiotics in this context
Gene/cell/RNAi/CRISPR functional-genomics screens; human omics signatures	Not Available	No human transcriptomic/proteomic/metabolomic disease datasets identified

The scientifically meaningful "genetics" of this disease reside in the **pathogen genome**—the *F. necrophorum* leukotoxin operon (*lktA*) and the hemagglutinin-related protein gene, both confirmed by PCR and Southern hybridization in virulent isolates ([PMID: 24590666](#)).

Section-by-Section Digest

- **1. Disease information:** MONDO:0015306; synonyms include necrobacillosis (human), postanginal sepsis, human necrobacillosis, "the forgotten disease." Information is derived from aggregated case series/systematic reviews and individual case reports (there is no large EHR-based cohort; the BATTLE registry, [PMID: 33091703](#), is assembling disease-specific data).
- **2. Etiology:** Infectious (predominantly *F. necrophorum*; expanding to streptococci and *S. aureus*). Risk factors: antecedent pharyngotonsillitis/peritonsillar abscess, young age, male sex, and inappropriate/restricted antibiotic use for sore throat. Smoking is an established risk factor for peritonsillar abscess (the main precursor). No genetic risk/protective factors; no established gene-environment interaction.
- **3. Phenotypes:** Fever (92%), sore throat, neck pain/swelling, rigors, dyspnea, pleuritic chest pain, hemoptysis; laboratory: thrombocytopenia (~75%), elevated CRP, leukocytosis. Onset is acute in previously healthy young people; severity moderate-to-severe; course rapidly progressive then resolving with treatment.
- **4. Genetic/molecular:** Not Applicable (see table above).
- **5. Environmental/infectious agents:** *Fusobacterium necrophorum* subsp. *necrophorum* (biotype A, more virulent) and subsp. *funduliforme*; viridans streptococci; MRSA (Lemierre-like). No toxin/pollution etiology.
- **6. Mechanism:** See causal chain above.
- **7. Anatomy:** oropharynx/tonsils → internal jugular vein → lungs (dominant), with intracranial-sinus, pleural, joint, and orbital extension.
- **8. Temporal:** acute/subacute onset days after pharyngitis; rapid progression; self-limited with treatment; no chronic/relapsing phase.
- **9. Inheritance/population:** No inheritance. Incidence ~1–10/million/year (rising); median age ~15–20 years; male predominance; global distribution.
- **10. Diagnostics:** contrast-enhanced neck/chest CT (IJV thrombosis + emboli) + anaerobic blood cultures; molecular (16S rDNA, targeted NGS/metagenomics); differential includes other deep-neck-space infections, viral pharyngitis, infectious mononucleosis, and non-septic venous thrombosis.
- **11. Prognosis:** ~5% mortality (up to 80% historically); high acute morbidity; near-complete recovery with treatment; prognostic factors—diagnostic delay, septic shock, and extent of embolization.

- **12. Treatment:** prolonged anaerobe-active antibiotics (beta-lactam/BLI or 3rd-gen cephalosporin + metronidazole, or carbapenem) + source control; anticoagulation controversial; IJV ligation/excision as salvage.
 - **13. Prevention:** secondary (early recognition + appropriate antibiotics); cost-effective throat-swab screening modeled but not implemented; no human vaccine.
 - **14. Other species:** major veterinary disease (bovine liver abscess, footrot; necrobacillosis in ruminants and wildlife); zoonotic transmission not a feature of human LS.
 - **15. Model organisms:** no dedicated rodent LS model; mechanistic work uses bovine PMN/leukocyte in-vitro systems and cattle field trials; a *F. necrophorum* bacterin vaccine is efficacious in cattle.
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Evidence Base

PMID	Title (abbrev.)	Role in this report
42127655	<i>Pediatric Lemierre's syndrome: A systematic review</i>	Dominant pathogen (78.3%), CT utility (89.5%), fever (92%), complications 4.2%, anticoagulation 62.2%
41517763	<i>Early diagnosis using targeted NGS + metagenomics</i>	Definition; molecular diagnostics for culture-negative cases
31843654	<i>Invasive F. necrophorum incl. LS: Swedish 8-year study</i>	Rising incidence (2.9→5.0/M/yr); severity metrics; median age 20
42576194	<i>Atypical LS in a child with nephrotic syndrome</i>	Incidence estimate 3.6/million
40265142	<i>A subtle presentation: Lemierre</i>	Historical 80% / modern ~5% mortality; 7–15 day course
8711893	<i>F. necrophorum virulence factors</i>	Leukotoxin + endotoxin as principal virulence factors
12117974	<i>Leukotoxin induces activation/apoptosis of bovine leukocytes</i>	Mechanism of immune evasion (in-vitro/ ruminant)
16701574	<i>F. necrophorum infections in animals</i>	Subspecies virulence; veterinary disease; "mechanism not well defined"
39840169	<i>Early Radiographic Warning Signs</i>	Confirmatory diagnostic standard; 4-day rapid progression
40224244	<i>LS in an immunocompetent patient</i>	First-line antibiotic regimen; contrast-enhanced CT
32909436	<i>Anticoagulation strategies: systematic review</i>	Anticoagulation controversy
42141231	<i>Otogenic pediatric septic thrombophlebitis</i>	Thrombosis distribution; 78% recanalization
20570017	<i>"A forgotten disease"</i>	Oropharyngeal source + pulmonary emboli; carbapenems
38363930	<i>Lemierre-like syndrome after MRSA soft-tissue infection</i>	Expanding etiologic spectrum; ~16% LLS mortality

PMID	Title (abbrev.)	Role in this report
42286749	<i>Incomplete Lemierre's syndrome</i>	Clinical variant definition
42517948	<i>IJV excision in uncontrolled LS</i>	Odontogenic/non-Fusobacterium cases; salvage surgery; DIC/ARDS
22886057	<i>Cost-effectiveness of throat-swab screening</i>	Secondary prevention economics (\$8,795/QALY)
16363327	<i>Vaccination against <i>F. necrophorum</i> in feedlot cattle</i>	Bacterin efficacy (liver abscess OR 0.27; footrot OR 0.18)
24590666	<i>Fusobacterium isolates from white-tailed deer</i>	Leukotoxin operon confirmation; wildlife reservoir
12109395	<i>Postanginal sepsis in South West Peninsula</i>	Otogenic sources; antibiotic-use link to rising incidence
39257960	<i>LS with extensive thrombosis</i>	Surgical hierarchy; anticoagulation for extensive thrombosis
33091703	<i>BATTLE registry rationale</i>	Registry to address evidence gaps; broadened definition
28260599	<i>Peritonsillar abscess microbiology & risk factors</i>	Smoking as PTA risk factor; FN prevalence in the precursor lesion
42547547	<i>Microbiological spectrum of peritonsillar abscesses</i>	FN more common in adolescents; antibiotic susceptibility

Limitations and Knowledge Gaps

- 1. Mechanistic gap (human thrombogenesis):** The step linking *F. necrophorum* virulence factors to actual IJV thrombus initiation in humans is inferred from in-vitro and ruminant data; direct human evidence of platelet aggregation/endothelial thrombus formation is lacking (Finding 12).
- 2. No human omics:** No transcriptomic, proteomic, metabolomic, single-cell, or spatial datasets exist for human LS, limiting molecular annotation and biomarker discovery.

3. **Evidence quality:** The literature is dominated by case reports, retrospective series, and systematic reviews of heterogeneous cases; there are no randomized trials for the central management controversy (anticoagulation).
 4. **Definitional drift:** "Lemierre-like" and "incomplete" variants broaden the disease boundary and complicate epidemiologic comparisons; incidence estimates vary (1–10/ million/year).
 5. **Underdiagnosis/publication bias:** As "the forgotten disease," LS may be under-recognized (biasing incidence downward) while severe/unusual cases are over-represented in the literature (biasing severity upward).
 6. **No dedicated small-animal model:** Mechanistic study relies on cattle/bovine leukocyte systems, which may not fully recapitulate human neck-vein thrombophlebitis.
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Proposed Follow-up Actions

1. **Contribute to / mine the BATTLE registry** ([PMID: 33091703](#)) for prospective, standardized data on presentation, management, and anticoagulation outcomes.
 2. **Design a pragmatic RCT of adjunctive anticoagulation** in confirmed LS, stratified by extent of thrombosis and intracranial-sinus involvement, to resolve the central management controversy.
 3. **Human mechanistic studies:** ex-vivo assays testing *F. necrophorum* leukotoxin, hemagglutinin, and outer-membrane factors against human platelets and jugular-vein endothelial cells to directly test the thrombus-initiation hypothesis (closing the Finding 12 gap).
 4. **Molecular diagnostics validation:** prospective evaluation of targeted NGS/16S rDNA turnaround time and sensitivity versus anaerobic culture, to shorten diagnostic delay (the key modifiable prognostic factor).
 5. **Prospective evaluation of throat-swab screening** in 15–24-year-olds for *F. necrophorum* pharyngitis to test the modeled cost-effectiveness ([PMID: 22886057](#)) as a secondary-prevention strategy.
 6. **Comparative-biology translation:** leverage the efficacious bovine *F. necrophorum* bacterin ([PMID: 16363327](#)) and leukotoxin biology to explore antitoxin/antivirulence adjuncts, while recognizing the low human incidence limits vaccine feasibility.
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Suggested Ontology Term Set (for knowledge-base ingestion)

- **Disease:** MONDO:0015306 (Lemierre syndrome)
- **Pathogen / animal (NCBI Taxon):** *Fusobacterium necrophorum* txid859; *Bos taurus* txid9913; *Odocoileus virginianus* txid9874
- **Anatomy (UBERON):** internal jugular vein UBERON:0001584; oropharynx UBERON:0001729; palatine tonsil UBERON:0002373; lung UBERON:0002048; sigmoid sinus UBERON:0006459; cavernous sinus UBERON:0004024
- **Phenotypes (HP):** Fever HP:0001945; Sepsis HP:0100806; Thrombocytopenia HP:0001873; Septic shock HP:0031273; Hemoptysis HP:0002105; Chest pain HP:0100749
- **Processes (GO):** inflammatory response GO:0006954; apoptotic process GO:0006915; blood coagulation GO:0007596; toxin activity GO:0090729
- **Cells (CL):** neutrophil CL:0000775; endothelial cell CL:0000115; platelet CL:0000233
- **Chemicals (CHEBI):** lipopolysaccharide CHEBI:16412; metronidazole CHEBI:6077
- **Interventions (NCIT-type):** antibacterial therapy; metronidazole; carbapenem; anticoagulant therapy; abscess drainage; internal jugular vein ligation

Evidence source types across this report: predominantly human clinical (case series, systematic reviews, registries); in-vitro/model-organism for virulence mechanism (bovine leukocytes, cattle field trials); computational/economic for prevention modeling. No germline-genetic or human-omics evidence exists for this disease.